P-192 rational graph kernel in quartic equations

Shared bits -> two complete addition tracks

Constructor inputs: curve, G, Q and r; target logarithm absent

Quartic system -> rational graph subgroup

P-192:4032 variables,17088 equations; projection has r points

Toy projection -> coordinate algebra

32 lifted assignments ->19 points; quotient evaluation remains open

Derived construction. Given E/F_p:y^2=x^3+ax+b with p>3, nonzero G,Q of odd prime order r, and k=bitlength(r), construct equations for Gamma=<{(G,Q)}> using shared Boolean digits e_0,...,e_(k-1). Precompute [2^i]G and [2^i]Q. Two addition tracks accumulate the selected points using the same digits. Every projected output is ([j]G,[j]Q), where j is the integer represented by those digits. No discrete logarithm of Q is an input to the constructor. The committed input file contains p,r,a,b,G,Q only. [pcc:unverified]

Size. Each point uses normalized coordinates (x,y,o): finite points have o=0 and y^2=x^3+ax+b; the identity is (0,0,1). One addition gate introduces three output coordinates, five branch selectors and two slope/inverse auxiliaries. It has 44 equations of degree at most four. The two tracks use 2k gates and k shared Boolean constraints. Thus the complete system has 21k variables and 89k equations, maximum degree four. For P-192, k=192: 4032 variables and 17088 equations. All basepoint multiples are constants, computed in O(k) group operations; they are not unknown variables. [pcc:unverified]

Complete addition predicate. The five disjoint branches are identity first input; identity second input with first finite; finite cancellation; finite doubling with nonzero 2y; ordinary finite addition with nonzero x-coordinate difference. Boolean selectors sum to one. Since p>=5 and there are five selectors, their sum condition admits exactly one active selector. Active-branch equations enforce the appropriate inverse and slope, output coordinates and identity flags; inactive inverse/slope variables are zero. The largest degree is four, from a selector times the doubling slope equation or the identity-flagged curve equation. The published template is instantiated with each selected input (e*x_i,e*y_i,1-e), which is a valid normalized point whenever e is Boolean. [pcc:unverified]

Correctness of projection. For every digit assignment, induction along the tracks gives one valid output and one complete assignment of all branch and arithmetic auxiliaries. Conversely, the point and branch equations force every gate to be the true group addition: nonzero denominators are witnessed by inverses, while cancellation includes the doubling of a point with y=0. Thus every solution projects to ([j]G,[j]Q). Since 0<=j<2^k and r<=2^k, all r residues occur. Distinct residues give distinct first coordinates as curve points because G has exact order r. Therefore the projected point set is exactly Gamma, with r points. The lifted Boolean assignments have 2^k points; multiple assignments may have the same projected point. This is a written algebraic argument, with finite controls below. [pcc:unverified]

Coordinate-algebra meaning. The Boolean constraints split the lifted algebra into one F_p component per digit assignment; in each component the complete addition equations uniquely solve the remaining variables. Its dimension is 2^k. The subalgebra generated by the output coordinates consists of functions constant on equal projected points and has dimension r: normalized coordinates separate the r output points, and their maximal ideals are pairwise comaximal. This describes an exact finite-algebra projection. A small quartic presentation does not by itself give a small eliminated representation or an inexpensive quotient evaluator. [pcc:unverified]

P-192 checks. The constructor receives a full-width fixture Q as point coordinates from round50, without reading its public test logarithm. Six digit strings cover j=0,1,2,r-1,r,2^k-1. A separate checker uses independent scalar and point arithmetic to reconstruct every witness, checks the branch semantics, point outputs, variable/equation counts, and witness digest. In particular, j=r returns both accumulators to the identity through the complete cancellation branch. The symbolic template degree is computed from the actual 44 equations. These are correctness checks of the compact system; no P-192 elimination is performed. [pcc:unverified]

Exhaustive small control. On E/F_17:y^2=x^3+2x+2, G=(5,1), r=19, Q=7G, all 32 digit strings are evaluated. Their projected image contains exactly 19 points. Every one of the 361 pairs of curve points is checked against the addition predicate and independent arithmetic, covering all five branches. Altering either an output coordinate or a selector rejects 722 mutations. A separate order-two point on y^2=x^3+x checks the y=0 cancellation case, and a nonbinary digit is rejected by its Boolean equation. [pcc:unverified]

Exact toy elimination. Enumerate the 32 lifted control assignments, merge duplicate projected points, and build the resulting coordinate algebra by exact F_17 linear algebra. A 19-element monomial basis of maximum total degree three is found. The certificate supplies its evaluation matrix, inverse, and six 19-by-19 coordinate multiplication matrices. The independent checker validates all 361 basis evaluations, both inverse identities, six coordinate actions, 15 commutation identities, six field equations,

and the normalized curve equations. This is an explicit projected coordinate algebra of dimension 19. The implementation uses enumeration of 2^k assignments and is not a P-192 elimination algorithm. [pcc:unverified]

Exact F17 algebra dimensions from all32 control assignments; no statistical uncertainty.

Connection to the mixed kernel. Round51 decomposes it as the rational graph Gamma plus its pairing-orthogonal line in the p-eigenspace. This round supplies actual equations for Gamma from the public points, rather than assuming the scalar-field invariant premise of rounds48-49. The variables and equations are over F_p, with auxiliary Boolean variables and projection; they are not a degree-four rational function in scalar labels over F_r. No scalar-field derivative extractor follows from this presentation. The non-rational pairing block and the abelian-surface quotient evaluator remain separate construction tasks. [pcc:unverified]

Next construction. Add a precise symbolic representation of the orthogonal p-eigenspace line and investigate elimination or invariant arithmetic that uses the compact representation directly. Any gain must account for handling its r-point coordinate algebra rather than infer efficiency from input degree alone. The other active native scalar-comparison task T-1888 was inspected and left untouched.